

Practical 10

08 April 2026

Questions

1. Obtain the consensus phylogenetic tree for the following two sets of sequences:

Set 1: tim.dat

Set 2: tim-hemo.dat

Hint:

- (i) Multiple sequence alignment using [MAFFT](#)
- (ii) Save in Phylip format
- (iii) Install [Phylip](#) (windows) / [Phylip](#) (Mac OS)
- (iv) Bootstrapping (Seqboot program)
- (v) Maximum likelihood method (proml program)
- (vi) Consensus tree (Consens program)
- (vii) Use [MEGA-X](#) to view the tree
- (viii) NJ and UPGMA methods (protdist and neighbor programs)

Refer to practical 10.ppt for detailed steps.

2. Obtain the weight matrix for the following sequences

```
MVLSPADKTNVKGKVGAGHAGEYGAAAW
MKRLPADPPCVKGKVKAKAGDYGATTW
MALSAADKTNVKSQVGGHAGEYGAATS
MVLSAADKTNVKSQAGGNAGEWWAAAW
MVLSAADKTNVKSQVLANAGEFGAAAW
ALLPIRTTYHKKCASGHIPEEKDLNNV
DEASSLKGHHIKKLEADALLIPLSASS
```

Deadline: 14 April 2026